Program: **Diploma in Mechatronics**	
Course Code: **6439**	Course Title: **Simulation Lab**
Semester: **6**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L:0, T:0, P:3)**	Periods per semester: **45**

Course Objectives:

- To simulate mathematical and functional aspects of analog electronic, digital electronic and power electronic circuits using the available numerical computing software. (eg: MATLAB/SIMULINK, SCILAB/XCOS, GNU OCTAVE)

Course Prerequisites:

Topic	Course code	Course name	Semester
Working of diode and transistor	2031	Fundamentals of Electrical and Electronics Engineering	2
Combinational logic circuits	3048	Digital Electronics Lab	3
Controlled rectifier circuits		Automation and Control Lab	5

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Develop programs to perform basic matrix operations, solve equations and plot basic signals	9	Applying
CO2	Develop simulation models to demonstrate the operation of electronic circuits	9	Applying
CO3	Develop simulation models to illustrate the operation of controlled rectifier circuits	9	Applying
CO4	Develop simulation models to implement combinational logic circuits	9	Applying
	Open Ended Project	3	
	Lab Exam	6	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3	3	3	3			
CO3	3	3	3	3			
CO4	3	3	3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	**Develop program to perform basic matrix operations, solve equations and plot basic signals**		
M1.01	Perform basic matrix operations – addition, subtraction, multiplication, division and inverse	1.5	Applying
M1.02	Solution of linear constant coefficient differential equation	1.5	Applying
M1.03	Determine roots of a polynomial	1.5	Applying
M1.04	Generation of various signals and sequences such as Step, Square, Saw tooth, Triangular, Sinusoidal, Ramp signal	3	Applying
M1.05	Plot different types of plots 3D, surface plot, polar plot	1.5	Applying
CO2	**Develop simulation models to demonstrate the operation of electronic circuits**		
M2.01	Simulate VI characteristics of diode.	1.5	Applying
M2.02	Simulate input and output characteristics of npn transistor.	3	Applying
M2.03	Simulate half wave rectifier with filter	1.5	Applying
M2.04	Simulate full wave rectifier with filter	1.5	Applying
M2.05	Simulate bridge rectifier with filter	1.5	Applying

	Lab Exam – I	3	

CO3	**Develop simulation models to illustrate the operation of controlled rectifier circuits**		
M3.01	Simulate single phase half-wave controlled rectifier with R and RL load	3	Applying
M3.02	Simulate single phase full-wave controlled rectifier with R and RL load	3	Applying
M3.03	Simulate single phase bridge controlled rectifier with R and RL load	3	Applying
CO4	**Develop simulation models to implement combinational logic circuits**		
M4.01	Simulate AND, OR, NAND, NOR, XOR and NOT Gates	1.5	Applying
M4.02	Develop simulation models for full adder and full subtractor	1.5	Applying
M4.03	Develop simulation models for multiplexer and demultiplexer	3	Applying
M4.04	Develop simulation models for D, T and JK flipflop	3	Applying
	Lab Exam II	3	

Open Ended Project

Students have to do open ended project during the course for the purpose of continuous evaluation. Open ended project is not for End Semester Examination, but compulsory to be included in Continuous Internal Evaluation. Students can do open-ended project as a group of 3-5. This open ended project shall be included in the bonafide record.

Text / Reference:

T/R	Book Title/Author
T1	AgamkumarTyagi, MATLAB and Simulink for Engineers, Oxford Higher Education,2011
T2	Anil Kumar Verma ,Scilab A Beginner'S Approach , Cengage India, 2021
R1	Cleve Moler, Experiments with MATLAB ,2011
R2	Sandeep Nagar, Introduction to Scilab: For Engineers and Scientists Kindle Edition, Apress, 2017
R3	Bansal/Goel/Sharma, MATLAB and its applications in engineering, Pearson, 2016

R4	Rohan Verm, NUMERICAL METHODS KIT: FOR MATLAB, SCILAB AND OCTAVE USERS , Notion Press, 2020

Online Resources:

Sl.No	Website Link
1	https://www.mathworks.com/content/dam/mathworks/mathworks-dot-com/moler/exm/book.pdf
2	https://www.mccormick.northwestern.edu/documents/students/undergraduate/introduction-to-matlab.pdf
3	https://www.scilab.org
4	https://scilab.in/spoken-tutorial
5	https://scilab.in/lab_migration/generate_lab/125/1
6	https://www.gnu.org/software/octave/
7	https://octave.org/octave.pdf